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MA5950: Mathematical Finance

Week 2 Assignment

Name:Rahul kumar
Roll Number:MA25C034

A1. St. Petersburg Paradox

Question. Explain, in 2–3 sentences, why the St. Petersburg Paradox demonstrates that maximizing expected monetary value cannot be the correct description of rational decision-making.

Solution

The **St. Petersburg Paradox** gives a gamble whose **expected monetary value is theoretically infinite**, yet a rational person would not be willing to pay an infinite amount to participate in it. Therefore, people do not make decisions based solely on expected monetary value; instead, they evaluate uncertain monetary outcomes according to their **utility**, which reflects diminishing marginal utility of wealth and risk preferences.

Question(A2).

For each utility function below, state whether the individual is risk-averse, risk-neutral, or risk-seeking, and justify your answer using the sign of $u''(W)$.

(a) $u(W) = 5W + 3$

(b) $u(W) = \sqrt{W}$

(c) $u(W) = W^2$

(d) $u(W) = -e^{-2W}$

Solution

The risk attitude of an individual can be determined from the second derivative of the utility function:

$$\boxed{\begin{array}{lcl} u''(W) < 0 & \Longrightarrow & \text{Risk-averse,} \\ u''(W) = 0 & \Longrightarrow & \text{Risk-neutral,} \\ u''(W) > 0 & \Longrightarrow & \text{Risk-seeking.} \end{array}}$$

(a) $u(W) = 5W + 3$

The first derivative is

$$u'(W) = 5.$$

The second derivative is

$$u''(W) = 0.$$

Since

$$u''(W) = 0,$$

the utility function is linear. Therefore, the individual is **risk-neutral**.

$$\boxed{\text{Risk-neutral}}$$

(b) $u(W) = \sqrt{W}$

Rewrite the utility function as

$$u(W) = W^{1/2}.$$

The first derivative is

$$u'(W) = \frac{1}{2}W^{-1/2} = \frac{1}{2\sqrt{W}}.$$

The second derivative is

$$u''(W) = -\frac{1}{4}W^{-3/2} = -\frac{1}{4W^{3/2}}.$$

For $W > 0$,

$$u''(W) < 0.$$

Therefore, the utility function is concave and the individual is **risk-averse**.

$$\boxed{\text{Risk-averse}}$$

(c) $u(W) = W^2$

The first derivative is

$$u'(W) = 2W.$$

The second derivative is

$$u''(W) = 2.$$

Since

$$u''(W) = 2 > 0,$$

the utility function is convex. Therefore, the individual is **risk-seeking**.

$$\boxed{\text{Risk-seeking}}$$

(d) $u(W) = -e^{-2W}$

The first derivative is

$$u'(W) = -\left(-2e^{-2W}\right) = 2e^{-2W}.$$

The second derivative is

$$u''(W) = 2(-2)e^{-2W} = -4e^{-2W}.$$

Since

$$e^{-2W} > 0,$$

we have

$$u''(W) = -4e^{-2W} < 0.$$

Therefore, the utility function is concave and the individual is **risk-averse**.

$$\boxed{\text{Risk-averse}}$$

Question(A3):

A3. State Jensen's Inequality for a concave function. What does the inequality look like instead for a convex utility function, and what risk attitude does that correspond to?

Solution

Let X be a random variable and let $u(\cdot)$ be a utility function.

1. Jensen's Inequality for a Concave Function

If $u(x)$ is a **concave** function, Jensen's Inequality states that

$$\boxed{u\left(E[X]\right) \geq E[u(X)]}$$

That is,

$$\boxed{u(\text{expected value}) \geq \text{expected utility}}$$

For a **strictly concave** function, if X is non-constant, then

$$u(E[X]) > E[u(X)].$$

A concave utility function satisfies

$$u''(x) < 0.$$

This represents **risk aversion**.

The intuition is that a risk-averse individual prefers receiving a certain amount equal to the expected value rather than accepting the risky prospect. Thus,

$$\boxed{u(E[X]) \geq E[u(X)]}$$

2. Jensen's Inequality for a Convex Function

If $u(x)$ is a **convex** function, Jensen's Inequality is reversed:

$$\boxed{u(E[X]) \leq E[u(X)]}$$

or equivalently,

$$\boxed{E[u(X)] \geq u(E[X])}$$

For a strictly convex utility function,

$$u''(x) > 0,$$

and for a non-constant random variable X,

$$E[u(X)] > u(E[X]).$$

3. Risk Attitude

A **convex utility function corresponds to risk-seeking behavior**.

A risk-seeking individual prefers the risky prospect to receiving its expected value with certainty. Therefore,

$$\boxed{E[u(X)] > u(E[X])}$$

4. Comparison

Utility Function	Jensen's Inequality	Risk Attitude
Concave, $u''(x) < 0$	$u(E[X]) \geq E[u(X)]$	Risk-averse
Linear, $u''(x) = 0$	$u(E[X]) = E[u(X)]$	Risk-neutral
Convex, $u''(x) > 0$	$u(E[X]) \leq E[u(X)]$	Risk-seeking

Conclusion

The key relationship is:

$$\boxed{\begin{array}{lllll} u''(x) < 0 & \Longrightarrow & \text{Concave utility} & \Longrightarrow & \text{Risk-averse}, \\ u''(x) = 0 & \Longrightarrow & \text{Linear utility} & \Longrightarrow & \text{Risk-neutral}, \\ u''(x) > 0 & \Longrightarrow & \text{Convex utility} & \Longrightarrow & \text{Risk-seeking}. \end{array}}$$

Question(A4):

A4. True or false, with justification:

"A risk-neutral individual would never purchase insurance."

Solution

Answer: False.

A risk-neutral individual **may purchase insurance**, even though they do not dislike risk itself.

For a risk-neutral individual, the utility function is linear:

$$u(W) = W,$$

so that

$$E[u(W)] = u(E[W]).$$

Suppose an individual has initial wealth W and faces a potential loss L with probability p.

1. Expected Wealth Without Insurance

Without insurance, the individual's wealth is

$$\begin{cases} W, & \text{with probability } 1-p, \\ W-L, & \text{with probability } p. \end{cases}$$

Therefore, expected wealth is

$$E[W_{\text{uninsured}}] = (1-p)W + p(W-L).$$

Simplifying,

$$E[W_{\text{uninsured}}] = W - pL.$$

Thus, the **actuarially fair insurance premium** is

$$\boxed{P = pL}.$$

2. Expected Wealth With Fair Insurance

If the individual purchases insurance at the actuarially fair premium

$$P = pL,$$

then their wealth after paying the premium is

$$W - P = W - pL.$$

Hence,

$$E[W_{\text{uninsured}}] = W_{\text{insured}}.$$

For a risk-neutral individual,

$$E[u(W_{\text{uninsured}})] = u(W - pL) = u(W_{\text{insured}}).$$

Therefore, the individual is **indifferent** between buying and not buying actuarially fair insurance.

3. Effect of the Insurance Premium

Consider three cases.

Case 1: Premium is less than the expected loss

If

$$P < pL,$$

then buying insurance increases expected wealth. Therefore, the risk-neutral individual would prefer to buy insurance.

Case 2: Premium equals the expected loss

If

$$P = pL,$$

then

$$W - P = W - pL.$$

The individual is indifferent between buying and not buying insurance.

Case 3: Premium is greater than the expected loss

If

$$P > pL,$$

then buying insurance decreases expected wealth. Therefore, the risk-neutral individual would prefer not to buy insurance.

Insurance Premium	Risk-Neutral Decision
$P < pL$	Buy insurance
$P = pL$	Indifferent
$P > pL$	Do not buy

A risk-neutral individual **can purchase insurance**, but they will do so only when it does not reduce their expected wealth. At an actuarially fair premium, they are indifferent between purchasing and not purchasing insurance.

Question(A5):

A5. Utility is only meaningful up to a positive affine transformation:

$$u(W) \quad \text{and} \quad v(W) = c + d \cdot u(W), \qquad d > 0,$$

represent identical preferences over lotteries. Explain briefly why $A(W)$ and $R(W)$ computed from u and from v must therefore be exactly equal.

Solution

We show that the **absolute risk aversion** $A(W)$ and the **relative risk aversion** $R(W)$ are unchanged under a positive affine transformation of utility.

Suppose

$$v(W) = c + d\,u(W), \qquad d > 0.$$

The absolute risk aversion is defined as

$$\boxed{A(W) = -\frac{u''(W)}{u'(W)}}$$

and the relative risk aversion is defined as

$$\boxed{R(W) = -W\frac{u''(W)}{u'(W)}}.$$

1. Absolute Risk Aversion

Since

$$v(W) = c + d\,u(W),$$

the first derivative is

$$v'(W) = d\,u'(W),$$

and the second derivative is

$$v''(W) = d\,u''(W).$$

Therefore, the absolute risk aversion computed using v is

$$A_v(W) = -\frac{v''(W)}{v'(W)}.$$

Substituting the derivatives of v gives

$$A_v(W) = -\frac{d\,u''(W)}{d\,u'(W)}.$$

Since $d > 0$, the factor d cancels:

$$\boxed{A_v(W) = -\frac{u''(W)}{u'(W)} = A_u(W)}.$$

Hence,

$$\boxed{A_v(W) = A_u(W)}.$$

2. Relative Risk Aversion

Similarly, the relative risk aversion computed using v is

$$R_v(W) = -W\frac{v''(W)}{v'(W)}.$$

Substituting

$$v'(W) = d\,u'(W), \qquad v''(W) = d\,u''(W),$$

we obtain

$$R_v(W) = -W\frac{d\,u''(W)}{d\,u'(W)}.$$

Again, d cancels:

$$\boxed{R_v(W) = -W\frac{u''(W)}{u'(W)} = R_u(W)}.$$

Therefore,

$$\boxed{R_v(W) = R_u(W)}.$$

Question(B1):

B1. For

$$u(W) = \sqrt{W},$$

a gamble pays Rs. 90 or Rs. 160 with equal probability. Compute the exact certainty equivalent CE and risk premium π by hand.

Solution

Given the utility function

$$u(W) = \sqrt{W},$$

and the gamble

$$W = \begin{cases} 90, & \text{with probability } \frac{1}{2}, \\ 160, & \text{with probability } \frac{1}{2}. \end{cases}$$

We need to find the **certainty equivalent** CE and the **risk premium** π.

1. Expected Utility

The expected utility of the gamble is

$$E[u(W)] = \frac{1}{2}u(90) + \frac{1}{2}u(160).$$

Since $u(W) = \sqrt{W}$,

$$E[u(W)] = \frac{1}{2}\sqrt{90} + \frac{1}{2}\sqrt{160}.$$

Simplifying the square roots,

$$\sqrt{90} = 3\sqrt{10},$$

and

$$\sqrt{160} = 4\sqrt{10}.$$

Therefore,

$$E[u(W)] = \frac{1}{2}(3\sqrt{10}) + \frac{1}{2}(4\sqrt{10})$$

$$= \frac{7\sqrt{10}}{2}.$$

Thus,

$$\boxed{E[u(W)] = \frac{7\sqrt{10}}{2}}.$$

2. Certainty Equivalent

The certainty equivalent CE is defined by

$$u(CE) = E[u(W)].$$

Since

$$u(W) = \sqrt{W},$$

we have

$$\sqrt{CE} = \frac{7\sqrt{10}}{2}.$$

Squaring both sides,

$$CE = \left(\frac{7\sqrt{10}}{2}\right)^2.$$

Hence,

$$CE = \frac{49(10)}{4} = \frac{490}{4} = \boxed{\frac{245}{2}}.$$

Therefore,

$$\boxed{CE = 122.5 \text{ rupees}}.$$

3. Expected Value of the Gamble

The expected monetary value is

$$E[W] = \frac{1}{2}(90) + \frac{1}{2}(160).$$

Thus,

$$E[W] = 45 + 80 = \boxed{125}.$$

4. Risk Premium

The risk premium is defined as

$$\pi = E[W] - CE.$$

Therefore,

$$\pi = 125 - \frac{245}{2}.$$

Writing 125 as $\frac{250}{2}$,

$$\pi = \frac{250}{2} - \frac{245}{2} = \boxed{\frac{5}{2}}.$$

Hence,

$$\boxed{\pi = 2.5 \text{ rupees}}.$$

Question(B2):

B2. Derive the Arrow–Pratt approximation

$$\boxed{\pi \approx \frac{1}{2} A(\bar{W})\sigma^2}$$

from a second-order Taylor expansion of $u(W)$ around $\bar{W}$, showing every step. This is the same derivation as in Lecture 5 – reproduce it in your own words to demonstrate that you understand each step, not just the final formula.

Solution

We derive the approximation

$$\boxed{\pi \approx \frac{1}{2} A(\bar{W})\sigma^2}$$

using a second-order Taylor expansion of the utility function around the expected wealth $\bar{W}$.

Step 1: Define Expected Wealth

Let W denote a random final wealth. Define its expected value as

$$\boxed{\bar{W} = E[W]}.$$

The variance of wealth is

$$\boxed{\sigma^2 = \text{Var}(W) = E[(W - \bar{W})^2]}.$$

The certainty equivalent CE is defined by

$$u(CE) = E[u(W)].$$

The risk premium is defined as

$$\boxed{\pi = E[W] - CE}.$$

Since

$$E[W] = \bar{W},$$

we have

$$\boxed{CE = \bar{W} - \pi}.$$

Therefore, the defining equation for the risk premium becomes

$$\boxed{u(\bar{W} - \pi) = E[u(W)]}.$$

Step 2: Taylor Expansion of $u(W)$ Around $\bar{W}$

Using a second-order Taylor expansion around $\bar{W}$,

$$u(W) \approx u(\bar{W}) + u'(\bar{W})(W - \bar{W}) + \frac{1}{2}u''(\bar{W})(W - \bar{W})^2.$$

Taking expectations on both sides,

$$E[u(W)] \approx E\left[u(\bar{W}) + u'(\bar{W})(W - \bar{W}) + \frac{1}{2}u''(\bar{W})(W - \bar{W})^2\right].$$

Using the linearity of expectation,

$$E[u(W)] \approx u(\bar{W}) + u'(\bar{W})E[W - \bar{W}] + \frac{1}{2}u''(\bar{W})E[(W - \bar{W})^2].$$

Now,

$$E[W - \bar{W}] = E[W] - \bar{W} = \bar{W} - \bar{W} = 0,$$

and

$$E[(W - \bar{W})^2] = \sigma^2.$$

Therefore,

$$\boxed{E[u(W)] \approx u(\bar{W}) + \frac{1}{2}u''(\bar{W})\sigma^2}.$$

Step 3: Taylor Expansion at the Certainty Equivalent

Recall that

$$CE = \bar{W} - \pi.$$

Therefore,

$$u(CE) = u(\bar{W} - \pi).$$

Expanding $u(\bar{W} - \pi)$ around $\bar{W}$,

$$u(\bar{W} - \pi) \approx u(\bar{W}) + u'(\bar{W})(-\pi) + \frac{1}{2}u''(\bar{W})(-\pi)^2.$$

Since

$$(-\pi)^2 = \pi^2,$$

we obtain

$$u(\bar{W} - \pi) \approx u(\bar{W}) - u'(\bar{W})\pi + \frac{1}{2}u''(\bar{W})\pi^2.$$

Because the risk premium π is assumed to be small, the second-order term π^2 is negligible compared with the first-order term. Hence,

$$\boxed{u(CE) \approx u(\bar{W}) - u'(\bar{W})\pi}.$$

Step 4: Use the Definition of the Certainty Equivalent

By definition,

$$u(CE) = E[u(W)].$$

Using the two approximations derived above,

$$u(\bar{W}) - u'(\bar{W})\pi \approx u(\bar{W}) + \frac{1}{2}u''(\bar{W})\sigma^2.$$

Canceling $u(\bar{W})$ from both sides gives

$$-u'(\bar{W})\pi \approx \frac{1}{2}u''(\bar{W})\sigma^2.$$

Therefore,

$$\pi \approx -\frac{u''(\bar{W})}{2u'(\bar{W})}\sigma^2.$$

Thus,

$$\boxed{\pi \approx -\frac{1}{2}\frac{u''(\bar{W})}{u'(\bar{W})}\sigma^2}.$$

Step 5: Introduce Arrow–Pratt Absolute Risk Aversion

The Arrow–Pratt coefficient of absolute risk aversion is defined as

$$\boxed{A(W) = -\frac{u''(W)}{u'(W)}}.$$

Evaluating at $\bar{W}$,

$$A(\bar{W}) = -\frac{u''(\bar{W})}{u'(\bar{W})}.$$

Substituting this into the expression for π gives

$$\boxed{\pi \approx \frac{1}{2}A(\bar{W})\sigma^2}.$$

Question(B3):

B3. Using the Arrow–Pratt approximation, estimate the risk premium for a log-utility investor

$$u(W) = \ln W$$

with

$$\bar{W} = \text{Rs. } 20,00,000$$

facing a risk with

$$\sigma = \text{Rs. } 3,00,000.$$

Solution

The Arrow–Pratt approximation for the risk premium is

$$\boxed{\pi \approx \frac{1}{2}A(\bar{W})\sigma^2}$$

where $A(W)$ is the Arrow–Pratt coefficient of absolute risk aversion.

Step 1: Find the Arrow–Pratt Coefficient

The absolute risk aversion coefficient is defined as

$$A(W) = -\frac{u''(W)}{u'(W)}.$$

For the log-utility function

$$u(W) = \ln W,$$

the first derivative is

$$u'(W) = \frac{1}{W},$$

and the second derivative is

$$u''(W) = -\frac{1}{W^2}.$$

Therefore,

$$A(W) = -\frac{-1/W^2}{1/W}.$$

Simplifying,

$$A(W) = \frac{1}{W}.$$

Hence,

$$\boxed{A(W) = \frac{1}{W}}.$$

At the expected wealth

$$\bar{W} = 20,00,000,$$

we have

$$A(\bar{W}) = \frac{1}{20,00,000}.$$

Therefore,

$$\boxed{A(\bar{W}) = 5 \times 10^{-7}}.$$

Step 2: Calculate the Variance

We are given

$$\sigma = 3,00,000.$$

Therefore,

$$\sigma^2 = (3,00,000)^2.$$

Hence,

$$\sigma^2 = 90,000,000,000 = 9 \times 10^{10}.$$

Thus,

$$\boxed{\sigma^2 = 9 \times 10^{10}}.$$

Step 3: Apply the Arrow–Pratt Approximation

Using

$$\pi \approx \frac{1}{2} A(\bar{W}) \sigma^2,$$

we obtain

$$\pi \approx \frac{1}{2} \left(\frac{1}{20,00,000} \right) (3,00,000)^2.$$

Substituting the value of σ^2,

$$\pi = \frac{1}{2} \left(\frac{90,000,000,000}{2,000,000} \right).$$

Since

$$\frac{90,000,000,000}{2,000,000} = 45,000,$$

we obtain

$$\pi = \frac{1}{2}(45,000) = 22,500.$$

Therefore,

$$\boxed{\pi \approx \text{Rs. } 22,500}.$$

Question(B4):

B4. An investor faces a gamble of ±Rs. 40,000 around a mean wealth of Rs. 10,00,000 (equal probability), and their measured certainty equivalent for this gamble is Rs. 9,99,200 (so $\pi = 800$). Using the Arrow–Pratt approximation, back out this investor's implied absolute risk aversion coefficient $A(\bar{W})$.

Solution

We are given:

$$\bar{W} = \text{Rs. } 10,00,000,$$

the gamble is ±Rs. 40,000 with equal probability, and

$$CE = \text{Rs. } 9,99,200.$$

Therefore, the risk premium is

$$\pi = \bar{W} - CE.$$

Hence,

$$\pi = 10,00,000 - 9,99,200 = \boxed{\text{Rs. } 800}.$$

Step 1: Arrow–Pratt Approximation

The Arrow–Pratt approximation for the risk premium is

$$\boxed{\pi \approx \frac{1}{2} A(\bar{W})\sigma^2}.$$

We need to solve for $A(\bar{W})$. Rearranging,

$$A(\bar{W}) \approx \frac{2\pi}{\sigma^2}.$$

Step 2: Calculate the Variance

The gamble is ±Rs. 40,000 around the mean wealth. Therefore,

$$W - \bar{W} = \begin{cases} 40,000, & \text{with probability } \frac{1}{2}, \\ -40,000, & \text{with probability } \frac{1}{2}. \end{cases}$$

The variance is

$$\sigma^2 = E[(W - \bar{W})^2].$$

Therefore,

$$\sigma^2 = \frac{1}{2}(40,000)^2 + \frac{1}{2}(-40,000)^2.$$

Since

$$(-40,000)^2 = (40,000)^2,$$

we obtain

$$\sigma^2 = (40,000)^2.$$

Thus,

$$\boxed{\sigma^2 = 1,600,000,000}$$

or equivalently,

$$\boxed{\sigma^2 = 1.6 \times 10^9}.$$

Step 3: Back Out Absolute Risk Aversion

Using

$$A(\bar{W}) \approx \frac{2\pi}{\sigma^2},$$

we obtain

$$A(\bar{W}) = \frac{2(800)}{(40,000)^2}.$$

Therefore,

$$A(\bar{W}) = \frac{1600}{1,600,000,000}.$$

Simplifying,

$$A(\bar{W}) = 10^{-6}.$$

Hence,

$$\boxed{A(\bar{W}) \approx 1 \times 10^{-6}}$$

per rupee.

Question(B5):

B5. Suppose the investor in B4 has power (CRRA) utility. Using your answer to B4 and the relationship

$$R(W) = W \cdot A(W),$$

find their implied relative risk-aversion parameter γ. What specific utility function does this correspond to?

Solution

From B4, we obtained

$$\boxed{A(\bar{W}) = 10^{-6}}.$$

The investor's mean wealth is

$$\boxed{\bar{W} = \text{Rs. } 10,00,000 = 10^6}.$$

Step 1: Relationship Between Absolute and Relative Risk Aversion

The relative risk-aversion coefficient is defined as

$$\boxed{R(W) = WA(W)}.$$

At the investor's mean wealth,

$$R(\bar{W}) = \bar{W}A(\bar{W}).$$

Substituting the values,

$$R(\bar{W}) = (10,00,000)(10^{-6}).$$

Since

$$10,00,000 = 10^6,$$

we obtain

$$R(\bar{W}) = 10^6 \times 10^{-6} = 1.$$

Therefore,

$$\boxed{R(\bar{W}) = 1}.$$

Step 2: Find the CRRA Parameter γ

For CRRA (Constant Relative Risk Aversion) utility,

$$\boxed{R(W) = \gamma}.$$

Therefore,

$$\boxed{\gamma = 1}.$$

Step 3: Determine the Specific Utility Function

The standard CRRA utility function is

$$u(W) = \begin{cases} \dfrac{W^{1-\gamma}}{1-\gamma}, & \gamma \neq 1, \\ \ln W, & \gamma = 1. \end{cases}$$

Since we found

$$\gamma = 1,$$

we are in the special logarithmic case.
Therefore, the investor's utility function is

$$\boxed{u(W) = \ln W}.$$

More generally, any positive affine transformation

$$u(W) = a + b \ln W, \qquad b > 0,$$

represents the same preferences.

Question(C1):

C1. Consider

$$u(W) = -\frac{1}{W}.$$

Compute $A(W)$ and $R(W)$ directly from this form (do not just quote the general CRRA formula). Confirm your answer matches the $\gamma = 2$ case of power utility from Lecture 6, and briefly explain why this must be so given Part A5.

Solution

Given the utility function

$$u(W) = -\frac{1}{W},$$

we need to compute the absolute risk aversion

$$\boxed{A(W) = -\frac{u''(W)}{u'(W)}}$$

and the relative risk aversion

$$\boxed{R(W) = -W\frac{u''(W)}{u'(W)}}.$$

We calculate these directly from the given utility function.

Step 1: First Derivative

Write the utility function as

$$u(W) = -W^{-1}.$$

Differentiating with respect to W,

$$u'(W) = (-1)(-1)W^{-2}.$$

Therefore,

$$\boxed{u'(W) = W^{-2} = \frac{1}{W^2}}.$$

Step 2: Second Derivative

Differentiating again,

$$u''(W) = -2W^{-3}.$$

Therefore,

$$\boxed{u''(W) = -\frac{2}{W^3}}.$$

Thus,

$$\boxed{u'(W) = \frac{1}{W^2}, \qquad u''(W) = -\frac{2}{W^3}}.$$

Step 3: Absolute Risk Aversion

The Arrow–Pratt coefficient of absolute risk aversion is

$$A(W) = -\frac{u''(W)}{u'(W)}.$$

Substituting the derivatives,

$$A(W) = -\frac{-2/W^3}{1/W^2}.$$

Therefore,

$$A(W) = \frac{2}{W^3} \times W^2 = \boxed{\frac{2}{W}}.$$

Hence,

$$\boxed{A(W) = \frac{2}{W}}.$$

Step 4: Relative Risk Aversion

The relative risk aversion is

$$R(W) = -W\frac{u''(W)}{u'(W)}.$$

Substituting the derivatives,

$$R(W) = -W\left(\frac{-2/W^3}{1/W^2}\right).$$

Simplifying the fraction,

$$\frac{-2/W^3}{1/W^2} = -\frac{2}{W}.$$

Therefore,

$$R(W) = -W\left(-\frac{2}{W}\right) = \boxed{2}.$$

Thus,

$$\boxed{R(W) = 2}.$$

Step 5: Comparison with the $\gamma = 2$ Power Utility

The standard CRRA power utility function is

$$u_\gamma(W) = \frac{W^{1-\gamma}}{1-\gamma}, \qquad \gamma \neq 1.$$

For

$$\gamma = 2,$$

we obtain

$$u_2(W) = \frac{W^{1-2}}{1-2}.$$

Therefore,

$$u_2(W) = \frac{W^{-1}}{-1} = -\frac{1}{W}.$$

Hence,

$$\boxed{u_2(W) = -\frac{1}{W}}.$$

This is exactly the utility function given in the question.

For CRRA utility, the relative risk-aversion parameter is

$$R(W) = \gamma.$$

Since we directly calculated

$$R(W) = 2,$$

we confirm that

$$\boxed{\gamma = 2}.$$

Step 6: Why Must This Agree with Part A5?

Part A5 established that utility functions related by a positive affine transformation

$$v(W) = c + d\,u(W), \qquad d > 0,$$

represent the same preferences and have exactly the same absolute and relative risk-aversion coefficients.

Therefore, even if the Lecture 6 power utility is written in a different but positively affine-equivalent form, it must produce the same values of $A(W)$ and $R(W)$.

In this case,

$$u(W) = -\frac{1}{W}$$

is exactly the $\gamma = 2$ CRRA utility under the standard normalization. Therefore,

$$\boxed{A(W) = \frac{2}{W}, \qquad R(W) = 2, \qquad \gamma = 2}.$$

Question(C2):

C2. For CARA utility, $R(W) = aW$ is increasing in wealth. Explain, in economic terms, what this implies about how a CARA investor's optimal portfolio share (fraction of wealth) invested in a risky asset should change as they become wealthier – and contrast this with what CRRA utility would imply instead.

Solution

For **CARA (Constant Absolute Risk Aversion)** utility, the absolute risk-aversion coefficient is constant:

$$\boxed{A(W) = a > 0}.$$

Therefore, relative risk aversion is

$$R(W) = WA(W).$$

Hence,

$$\boxed{R(W) = aW}$$

which is increasing in wealth.

1. Economic Interpretation for a CARA Investor

Relative risk aversion measures how strongly an investor dislikes risk relative to their wealth.

Since

$$R(W) = aW,$$

as wealth W increases,

$$R(W) \uparrow .$$

Thus, a CARA investor becomes **more relatively risk-averse** as they become wealthier.

However, it is important to distinguish between:

- the **dollar amount** invested in the risky asset, and
- the **fraction of wealth** invested in the risky asset.

Under CARA utility,

$$A(W) = a$$

is constant. For a given risky investment opportunity, this implies that the investor's optimal **dollar exposure** to the risky asset can remain approximately constant as wealth changes.

Let the optimal dollar investment in the risky asset be x^*. Then the portfolio share is

$$s^* = \frac{x^*}{W}.$$

If x^* remains approximately constant while wealth increases, then

$$s^* = \frac{x^*}{W}$$

must decrease.

Therefore,

As wealth increases, a CARA investor reduces the fraction of wealth invested in the risky asset.

Example

Suppose the optimal risky investment is Rs. 1 lakh.

If wealth is Rs. 10 lakh,

$$s^* = \frac{1}{10} = 10\%.$$

If wealth increases to Rs. 20 lakh while the optimal risky investment remains Rs. 1 lakh,

$$s^* = \frac{1}{20} = 5\%.$$

Thus, the **rupee amount** invested in the risky asset stays roughly constant, while the **percentage of wealth** invested in it falls.

2. Contrast with CRRA Utility

For **CRRA (Constant Relative Risk Aversion)** utility, relative risk aversion is constant:

$$\boxed{R(W) = \gamma}.$$

Therefore, as wealth increases,

$$R(W) = \gamma$$

does not change.
This means that the investor's attitude toward risk **relative to wealth remains constant**.
Consequently, for a given investment opportunity, the optimal **portfolio share** invested in the risky asset is independent of wealth:

$$\boxed{s^* = \text{constant}}.$$

For example, if the optimal risky-asset share is 20%, then

$$W = 10 \text{ lakh} \quad \Rightarrow \quad \text{risky investment} = 2 \text{ lakh},$$

while

$$W = 20 \text{ lakh} \quad \Rightarrow \quad \text{risky investment} = 4 \text{ lakh}.$$

The fraction remains

$$\boxed{20\%}.$$

	CARA	**CRRA**
$A(W)$	a	$\frac{\gamma}{W}$
$R(W)$	aW	γ
Relative risk aversion	Increases with wealth	Constant
Optimal risky dollar amount	Approximately constant	Increases with wealth
Optimal risky share	Decreases with wealth	Constant

For a CARA investor,

$$\boxed{\text{wealthier} \Rightarrow \text{more relatively risk-averse} \Rightarrow \text{smaller risky share.}}$$

For a CRRA investor,

$$\boxed{\text{wealthier} \Rightarrow \text{same relative risk aversion} \Rightarrow \text{same risky share.}}$$

Therefore, CARA investors tend to keep their **risky investment in rupee terms** roughly fixed, so its percentage of wealth falls as they become richer. In contrast, CRRA investors keep their **percentage allocation** roughly fixed, so their risky investment in rupee terms grows with wealth.

Question(C3):

C3. Prove the exact CARA result from Lecture 6: for

$$W = \bar{W} + \varepsilon,$$

where ε is a rupee-denominated risk independent of $\bar{W}$, show that

$$\boxed{CE = \bar{W} - \frac{1}{a} \ln E\left[e^{-a\varepsilon}\right]}$$

and hence that the risk premium π does not depend on $\bar{W}$ at all. Show every step.

Solution

We consider CARA utility

$$\boxed{u(W) = -e^{-aW}, \qquad a > 0.}$$

The wealth is given by

$$\boxed{W = \bar{W} + \varepsilon,}$$

where $\bar{W} = E[W]$ and ε represents the rupee-denominated risk.
We will derive the exact certainty equivalent and then show that the risk premium is independent of $\bar{W}$.

Step 1: Definition of the Certainty Equivalent

The certainty equivalent CE is the certain amount of wealth that gives the same utility as the expected utility of the risky wealth. Therefore,

$$u(CE) = E[u(W)].$$

Since CARA utility is

$$u(W) = -e^{-aW},$$

we have

$$u(CE) = -e^{-aCE}.$$

Thus,

$$-e^{-aCE} = E[-e^{-aW}].$$

Taking the negative sign outside the expectation,

$$-e^{-aCE} = -E[e^{-aW}].$$

Multiplying both sides by -1 gives

$$\boxed{e^{-aCE} = E[e^{-aW}]}.$$

Step 2: Substitute $W = \bar{W} + \varepsilon$

Since

$$W = \bar{W} + \varepsilon,$$

we have

$$e^{-aW} = e^{-a(\bar{W}+\varepsilon)}.$$

Using the property

$$e^{x+y} = e^x e^y,$$

we obtain

$$e^{-a(\bar{W}+\varepsilon)} = e^{-a\bar{W}} e^{-a\varepsilon}.$$

Therefore,

$$E[e^{-aW}] = E\left[e^{-a\bar{W}} e^{-a\varepsilon}\right].$$

Since $\bar{W}$ is the deterministic mean wealth, $e^{-a\bar{W}}$ is a constant and can be taken outside the expectation:

$$E[e^{-aW}] = e^{-a\bar{W}} E[e^{-a\varepsilon}].$$

Thus,

$$\boxed{e^{-aCE} = e^{-a\bar{W}} E[e^{-a\varepsilon}]}.$$

Step 3: Take Natural Logarithms

Taking the natural logarithm of both sides,

$$\ln(e^{-aCE}) = \ln\left(e^{-a\bar{W}} E[e^{-a\varepsilon}]\right).$$

The left-hand side becomes

$$-aCE.$$

Using

$$\ln(xy) = \ln x + \ln y,$$

the right-hand side becomes

$$\ln(e^{-a\bar{W}}) + \ln E[e^{-a\varepsilon}].$$

Therefore,

$$-aCE = -a\bar{W} + \ln E[e^{-a\varepsilon}].$$

Step 4: Solve for the Certainty Equivalent

Multiplying both sides by -1,

$$aCE = a\bar{W} - \ln E[e^{-a\varepsilon}].$$

Dividing by a,

$$CE = \bar{W} - \frac{1}{a} \ln E[e^{-a\varepsilon}].$$

Therefore,

$$\boxed{CE = \bar{W} - \frac{1}{a} \ln E[e^{-a\varepsilon}]}.$$

This is the exact CARA certainty equivalent.

Step 5: Derive the Risk Premium

The risk premium is defined as

$$\pi = E[W] - CE.$$

Since

$$W = \bar{W} + \varepsilon$$

and

$$\bar{W} = E[W],$$

we have

$$E[\varepsilon] = 0.$$

Therefore,

$$E[W] = E[\bar{W} + \varepsilon] = \bar{W} + E[\varepsilon] = \bar{W}.$$

Hence,

$$\pi = \bar{W} - CE.$$

Substituting the expression for CE,

$$\pi = \bar{W} - \left[\bar{W} - \frac{1}{a} \ln E[e^{-a\varepsilon}]\right].$$

The $\bar{W}$ terms cancel:

$$\pi = \bar{W} - \bar{W} + \frac{1}{a} \ln E[e^{-a\varepsilon}].$$

Thus,

$$\boxed{\pi = \frac{1}{a} \ln E[e^{-a\varepsilon}]}.$$

Step 6: Show Independence from Wealth

The final expression for the risk premium is

$$\boxed{\pi = \frac{1}{a} \ln E[e^{-a\varepsilon}]}.$$

Notice that $\bar{W}$ does not appear anywhere in this expression. Therefore,

$$\boxed{\frac{\partial \pi}{\partial \bar{W}} = 0}.$$

Hence, the risk premium is completely independent of the investor's mean wealth.

Economic Interpretation

For CARA utility,

$$u(W) = -e^{-aW},$$

the investor's risk premium depends only on the distribution of the absolute, rupee-denominated risk ε and the coefficient a.
Thus, if the same rupee-denominated gamble is added to investors with different levels of wealth, the exact risk premium is unchanged.
Therefore,

$$\boxed{\text{CARA} \quad \Longrightarrow \quad \text{Risk premium is independent of wealth.}}$$

This is the exact CARA result.

Question(C4):

C4. (5 marks) A CRRA investor has $\gamma = 4$. Facing a proportional gamble, their risk premium is

$$\frac{\pi}{\bar{W}} = 0.8\%$$

when

$$\bar{W} = \text{Rs. } 8,00,000.$$

Using only the exact CRRA scaling property from Lecture 6 (no new utility calculation), find π (in rupees) when

$$\bar{W} = \text{Rs. } 40,00,000.$$

Solution

We are given

$$\gamma = 4,$$

and for mean wealth

$$\bar{W}_1 = \text{Rs. } 8,00,000,$$

the proportional risk premium is

$$\frac{\pi_1}{\bar{W}_1} = 0.8\%.$$

We need to find the risk premium when

$$\bar{W}_2 = \text{Rs. } 40,00,000.$$

Step 1: Use the Exact CRRA Scaling Property

For CRRA utility, if a gamble is proportional to wealth, the risk premium scales proportionally with wealth. Therefore,

$$\boxed{\frac{\pi}{\bar{W}} = \text{constant}}.$$

Hence,

$$\frac{\pi_2}{\bar{W}_2} = \frac{\pi_1}{\bar{W}_1}.$$

Since

$$\frac{\pi_1}{\bar{W}_1} = 0.8\%,$$

we have

$$\frac{\pi_2}{\bar{W}_2} = 0.8\%.$$

Step 2: Calculate the New Risk Premium

We have

$$\bar{W}_2 = \text{Rs. } 40,00,000.$$

Therefore,

$$\pi_2 = 0.8\% \times 40,00,000.$$

Converting the percentage into decimal form,

$$0.8\% = \frac{0.8}{100} = 0.008.$$

Thus,

$$\pi_2 = 0.008(40,00,000).$$

Therefore,

$$\boxed{\pi_2 = \text{Rs. } 32,000}.$$

Step 3: Verification Using the Wealth Ratio

Notice that wealth increases by a factor of

$$\frac{40,00,000}{8,00,000} = 5.$$

Since CRRA implies proportional scaling,

$$\pi_2 = 5\pi_1.$$

First calculate π_1:

$$\pi_1 = 0.008(8,00,000) = \text{Rs. } 6,400.$$

Therefore,

$$\pi_2 = 5(6,400) = \boxed{\text{Rs. } 32,000}.$$

Question (E1)

E1. In Week 1, you learned that bonds are a fixed, senior claim while equities are a residual, uncapped claim with historically higher average returns. In Week 2, you learned that a risk-averse investor with concave utility values a risky payoff at less than its expected value, sacrificing a risk premium to avoid uncertainty. In a paragraph of 150–200 words, explain why a risk-averse investor with CRRA utility would rationally choose to hold both bonds and equities, rather than investing entirely in equities to chase the higher expected return. Your answer should reference:

1. (i)diminishing marginal utility of wealth,(ii)the risk premium concept from this week, and(iii) the fixed-vs-residual claim distinction from Week 1.

Solution

A risk-averse investor with CRRA utility would rationally hold a combination of bonds and equities because the investor cares about both expected return and risk. Under CRRA utility, marginal utility of wealth is diminishing: an additional rupee is more valuable when wealth is low than when wealth is high. Therefore, the investor dislikes large fluctuations in wealth and is willing to sacrifice some expected return, in the form of a **risk premium**, to reduce uncertainty. Equities are residual, uncapped claims, meaning equity holders receive what remains after fixed obligations to bondholders are paid. Consequently, equity returns can be high but are also more uncertain and can experience substantial losses. Bonds, in contrast, represent relatively fixed and senior claims, providing more predictable cash flows and greater protection against downside outcomes. Although equities historically offer a higher average return, this additional return is compensation for bearing additional risk. A risk-averse investor therefore does not simply maximize expected return; they maximize expected utility. Holding both bonds and equities provides diversification, reducing the variability of overall wealth while retaining some exposure to the higher expected returns of equities. Thus, the optimal portfolio balances the equity risk premium against the investor's aversion to risk.

Question (D1)

D1. **[Python]** Using your Lab code (or a fresh, self-contained script), consider a CARA investor with wealth

$$\bar{W} = \text{Rs. } 20 \text{ lakh}$$

facing a fixed additive gamble of ±Rs. 2 lakh with equal probability. Compute and plot the exact (root-found) risk premium π as a function of a, for a ranging from 0.01 to 0.20 (in units of 1/lakh). Comment on the shape of the curve: is it linear in a? Why or why not?

Solution

We are given

$$\bar{W} = 20 \text{ lakh},$$

and the gamble

$$\varepsilon = \begin{cases} +2 \text{ lakh}, & \text{with probability } \frac{1}{2}, \\ -2 \text{ lakh}, & \text{with probability } \frac{1}{2}. \end{cases}$$

Therefore, final wealth is

$$W = \begin{cases} 22 \text{ lakh}, & \text{with probability } \frac{1}{2}, \\ 18 \text{ lakh}, & \text{with probability } \frac{1}{2}. \end{cases}$$

For CARA utility,

$$\boxed{u(W) = -e^{-aW}}.$$

1. Exact Certainty Equivalent

The certainty equivalent satisfies

$$u(CE) = E[u(W)].$$

Therefore,

$$-e^{-aCE} = -\frac{1}{2}e^{-22a} - \frac{1}{2}e^{-18a}.$$

Multiplying both sides by -1 gives

$$e^{-aCE} = \frac{1}{2}e^{-22a} + \frac{1}{2}e^{-18a}.$$

Factor out e^{-20a}:

$$e^{-aCE} = e^{-20a}\left(\frac{1}{2}e^{-2a} + \frac{1}{2}e^{2a}\right).$$

Recall that

$$\cosh(x) = \frac{e^{x} + e^{-x}}{2}.$$

Therefore,

$$e^{-aCE} = e^{-20a}\cosh(2a).$$

Taking the natural logarithm of both sides,

$$-aCE = -20a + \ln(\cosh(2a)).$$

Dividing by $-a$,

$$\boxed{CE = 20 - \frac{1}{a}\ln(\cosh(2a))}.$$

Here, wealth and the certainty equivalent are measured in lakh rupees.

2. Exact Risk Premium

The risk premium is defined as

$$\pi = \bar{W} - CE.$$

Since

$$\bar{W} = 20,$$

we obtain

$$\pi = 20 - \left[20 - \frac{1}{a}\ln(\cosh(2a))\right].$$

The 20 terms cancel, giving

$$\boxed{\pi(a) = \frac{\ln(\cosh(2a))}{a}}.$$

Thus, the exact risk premium as a function of a is

$$\boxed{\pi(a) = \frac{\ln(\cosh(2a))}{a}}.$$

The risk premium is measured in lakh rupees.

3. Numerical Values

For selected values of a, the exact risk premium is approximately

a	$\pi(a)$ (lakh rupees)
0.01	0.019999
0.05	0.099917
0.10	0.198684
0.15	0.296696
0.20	0.389821

Thus, the risk premium increases as a increases.

4. Shape of the Curve

The exact relationship is

$$\pi(a) = \frac{\ln(\cosh(2a))}{a}.$$

This relationship is **not exactly linear** in a.
To understand the shape of the curve, consider small values of a. Using the Taylor expansion

$$\cosh(2a) = 1 + \frac{(2a)^2}{2!} + \frac{(2a)^4}{4!} + \cdots,$$

we obtain

$$\cosh(2a) = 1 + 2a^2 + \frac{2}{3}a^4 + \cdots.$$

Using the expansion

$$\ln(1+x) = x - \frac{x^2}{2} + \cdots,$$

we get

```python
import numpy as np
import matplotlib.pyplot as plt
from scipy.optimize import brentq


# ============================================================
# D1. CARA INVESTOR
# ============================================================

# Mean wealth: Rs. 20 lakh
mean_wealth = 20.0

# Gamble: +Rs. 2 lakh or -Rs. 2 lakh
gamble = 2.0


# ============================================================
# CARA UTILITY FUNCTION
# u(W) = -exp(-aW)
# ============================================================

def utility(W, a):
    return -np.exp(-a * W)


# ============================================================
# EXPECTED UTILITY OF THE GAMBLE
# ============================================================

def expected_utility(a):

    wealth_high = mean_wealth + gamble
    wealth_low = mean_wealth - gamble

    EU = (
        0.5 * utility(wealth_high, a)
        + 0.5 * utility(wealth_low, a)
    )

    return EU


# ============================================================
# CERTAINTY EQUIVALENT
#
# CE satisfies:
#
# u(CE) = E[u(W)]
#
# We find CE using numerical root-finding.
# ============================================================

def certainty_equivalent(a):

    EU = expected_utility(a)

    def equation(CE):
        return utility(CE, a) - EU

    CE = brentq(
        equation,
        mean_wealth - gamble,
        mean_wealth + gamble
    )

    return CE


# ============================================================
# RISK PREMIUM
#
# pi = Mean Wealth - Certainty Equivalent
# ============================================================

def risk_premium(a):

    CE = certainty_equivalent(a)

    return mean_wealth - CE


# ============================================================
# VALUES OF a
#
# a ranges from 0.01 to 0.20
# ============================================================

a_values = np.linspace(0.01, 0.20, 200)


# ============================================================
# CALCULATE RISK PREMIUM FOR EACH a
# ============================================================

risk_premiums = []

for a in a_values:

    pi = risk_premium(a)

    risk_premiums.append(pi)


# Convert list to NumPy array
risk_premiums = np.array(risk_premiums)


# ============================================================
# PRINT SOME RESULTS
# ============================================================

print(" a       CE (Rs. lakh)       Risk Premium (Rs. lakh)")
print("-" * 55)

for a in [0.01, 0.05, 0.10, 0.15, 0.20]:

    CE = certainty_equivalent(a)
    pi = risk_premium(a)

    print(f"{a:.2f}       {CE:.6f}              {pi:.6f}")


# ============================================================
```

Figure 1: Enter Caption

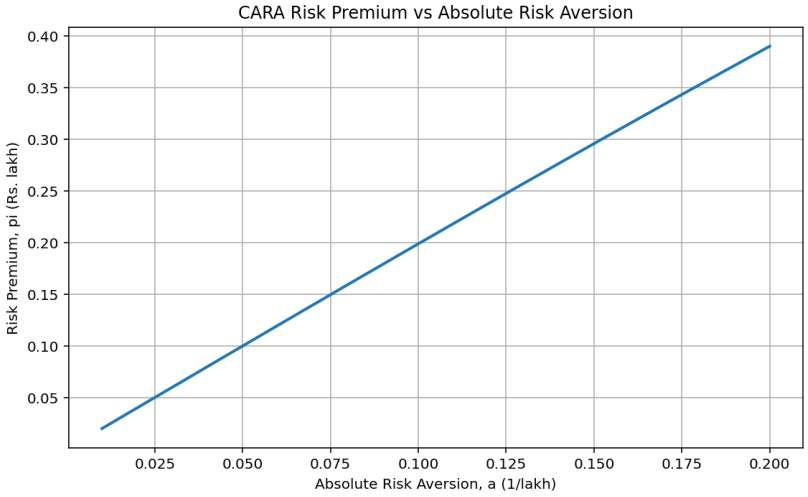


Figure 2: Enter Caption

$$\ln(\cosh(2a)) = 2a^2 - \frac{4}{3}a^4 + \cdots .$$

Therefore,

$$\pi(a) = \frac{2a^2 - \frac{4}{3}a^4 + \cdots}{a}.$$

Hence,

$$\boxed{\pi(a) = 2a - \frac{4}{3}a^3 + \cdots}.$$

For sufficiently small a,

$$\boxed{\pi(a) \approx 2a}.$$

Thus, the risk premium is approximately linear near $a = 0$.

However, as a increases, the higher-order terms become more important. Therefore, the exact curve deviates from the straight-line approximation and is **nonlinear**.

5. Conclusion

The exact risk premium is

$$\boxed{\pi(a) = \frac{\ln(\cosh(2a))}{a}}.$$

It increases with the coefficient of absolute risk aversion a. For small values of a, it is approximately

$$\boxed{\pi(a) \approx 2a},$$

which explains why the graph initially looks almost linear. Nevertheless, over the full range $0.01 \leq a \leq 0.20$, the exact relationship is nonlinear because of the higher-order terms in the Taylor expansion.

Therefore, the curve is **approximately linear for small a, but not exactly linear over the entire range**.

Question(D2):

D2. **[Python]** On the same axes as D1, plot the Arrow–Pratt approximation

$$\pi \approx \frac{1}{2}a\sigma^2$$

for the same range of a. For what range of a is the approximation within 5% of the exact value? Does this match what you would expect, given that the gamble size (Rs. 2 lakh, a 10% swing on Rs. 20 lakh) is held fixed while a increases?

Solution

From D1, the CARA investor has

$$\bar{W} = 20 \text{ lakh}$$

and faces the gamble

$$\varepsilon = \begin{cases} +2, & \text{with probability } \frac{1}{2}, \\ -2, & \text{with probability } \frac{1}{2}, \end{cases}$$

where all wealth figures are measured in lakh rupees.

1. Variance of the Gamble

Since the gamble has mean zero,

$$E[\varepsilon] = 0.$$

Its variance is

$$\sigma^2 = E[\varepsilon^2].$$

Therefore,

$$\sigma^2 = \frac{1}{2}(2^2) + \frac{1}{2}(-2)^2.$$

Hence,

$$\boxed{\sigma^2 = 4}.$$

2. Exact Risk Premium

From D1, the exact CARA risk premium is

$$\boxed{\pi_{\text{exact}}(a) = \frac{\ln(\cosh(2a))}{a}}.$$

3. Arrow–Pratt Approximation

The Arrow–Pratt approximation is

$$\pi \approx \frac{1}{2} A(\bar{W}) \sigma^2.$$

For CARA utility,

$$A(W) = a.$$

Therefore,

$$\pi_{\text{AP}} \approx \frac{1}{2} a \sigma^2.$$

Since

$$\sigma^2 = 4,$$

we obtain

$$\pi_{\text{AP}} = \frac{1}{2} a(4).$$

Thus,

$$\boxed{\pi_{\text{AP}}(a) = 2a}.$$

Therefore, the two curves to be plotted are

$$\boxed{\pi_{\text{exact}}(a) = \frac{\ln(\cosh(2a))}{a}}$$

and

$$\boxed{\pi_{\mathrm{AP}}(a) = 2a}.$$

The range of a is

$$0.01 \le a \le 0.20.$$

4. Relative Error

To determine when the Arrow–Pratt approximation is within 5% of the exact value, calculate the relative error:

$$\text{Relative Error} = \frac{|\pi_{\mathrm{AP}} - \pi_{\mathrm{exact}}|}{\pi_{\mathrm{exact}}}.$$

Substituting the two expressions,

$$\text{Relative Error} = \frac{\left|2a - \dfrac{\ln(\cosh(2a))}{a}\right|}{\dfrac{\ln(\cosh(2a))}{a}}.$$

For the range

$$0.01 \le a \le 0.20,$$

the approximation becomes progressively less accurate as a increases.
At the largest value,

$$a = 0.20,$$

the exact risk premium is

$$\pi_{\mathrm{exact}} = \frac{\ln(\cosh(0.4))}{0.2} \approx 0.38982 \text{ lakh}.$$

The Arrow–Pratt approximation is

$$\pi_{\mathrm{AP}} = 2(0.20) = 0.40 \text{ lakh}.$$

Therefore,

$$\text{Relative Error} = \frac{|0.40 - 0.38982|}{0.38982}.$$

Hence,

$$\text{Relative Error} \approx 0.0263 = 2.63\%.$$

Since

$$2.63\% < 5\%,$$

the approximation is within 5% over the entire specified range:

$$\boxed{0.01 \le a \le 0.20}.$$

5. Numerical Comparison

a	π_{exact}	π_{AP}	Relative Error
0.01	0.02000	0.02000	0.007%
0.05	0.09992	0.10000	0.167%
0.10	0.19868	0.20000	0.664%
0.15	0.29670	0.30000	1.49%
0.20	0.38982	0.40000	2.63%

Thus, the approximation remains within 5% throughout the required range.

6. Why Does This Make Sense?

The Arrow–Pratt approximation is obtained from a second-order Taylor expansion of the utility function. Therefore, it ignores terms of third and higher order.

The gamble is fixed at

$$\pm\text{Rs. } 2 \text{ lakh},$$

while mean wealth is

$$\text{Rs. } 20 \text{ lakh}.$$

Hence, the size of the gamble relative to wealth is

$$\frac{2}{20} = 0.10 = 10\%.$$

Thus, the gamble represents a 10% up-or-down movement in wealth.

For relatively small risks, the second-order Taylor approximation is very accurate. However, as a increases, the investor becomes more risk-averse and the higher-order terms become increasingly important.

Therefore, we expect the approximation to become gradually less accurate as a increases.

Indeed,

a	Relative Error
0.01	0.007%
0.05	0.167%
0.10	0.664%
0.15	1.49%
0.20	2.63%

The error increases with a, but it remains below 5% over the specified range.

7. Final Answer

The exact CARA risk premium is

$$\boxed{\pi_{\text{exact}}(a) = \frac{\ln(\cosh(2a))}{a}}$$

and the Arrow–Pratt approximation is

$$\boxed{\pi_{\text{AP}}(a) = 2a}.$$

The Arrow–Pratt approximation is within 5% of the exact risk premium for

$$\boxed{0.01 \le a \le 0.20}.$$

The result is consistent with the theory: because the gamble size is fixed, increasing a makes higher-order terms more important, so the Arrow–Pratt approximation becomes gradually less accurate. Nevertheless, for this particular 10% gamble and the specified range of a, the approximation remains very accurate.

```python
import numpy as np
import matplotlib.pyplot as plt

# ------------------------------------------------------------
# Parameters
# ------------------------------------------------------------

# Mean wealth = Rs. 20 lakh
W_bar = 20

# Gamble = +/- Rs. 2 lakh
b = 2

# Variance of the gamble
sigma_squared = b**2

# Range of a
a_values = np.linspace(0.01, 0.20, 200)


# ------------------------------------------------------------
# Exact CARA Risk Premium
# ------------------------------------------------------------

pi_exact = (
    np.log(np.cosh(a_values * b))
    / a_values
)


# ------------------------------------------------------------
# Arrow-Pratt Approximation
#
# pi в‰€ (1/2) A(W_bar) sigma^2
#
# For CARA: A(W_bar) = a
# ------------------------------------------------------------

pi_approx = (
    0.5
    * a_values
    * sigma_squared
)


# ------------------------------------------------------------
# Plot Both Curves
# ------------------------------------------------------------

plt.figure(figsize=(8, 5))

plt.plot(
    a_values,
    pi_exact,
    linewidth=2,
    label="Exact Risk Premium"
)

plt.plot(
    a_values,
    pi_approx,
    "--",
    linewidth=2,
    label="Arrow-Pratt Approximation"
)

plt.xlabel("Absolute Risk Aversion a (1/lakh)")
plt.ylabel("Risk Premium ПЂ (Rs. lakh)")

plt.title("Exact CARA Risk Premium vs Arrow-Pratt Approximation")

plt.legend()
plt.grid(True)

plt.tight_layout()
plt.show()


# ------------------------------------------------------------
# Calculate Relative Error
# ------------------------------------------------------------

relative_error = np.abs(
    pi_approx - pi_exact
) / pi_exact


# ------------------------------------------------------------
# Find values where error <= 5%
# ------------------------------------------------------------

valid = a_values[relative_error <= 0.05]

print("Range where approximation is within 5%:")

if len(valid) > 0:
    print(
        f"a = {valid[0]:.4f} to a = {valid[-1]:.4f}"
    )
else:
    print("No values in the specified range.")
```

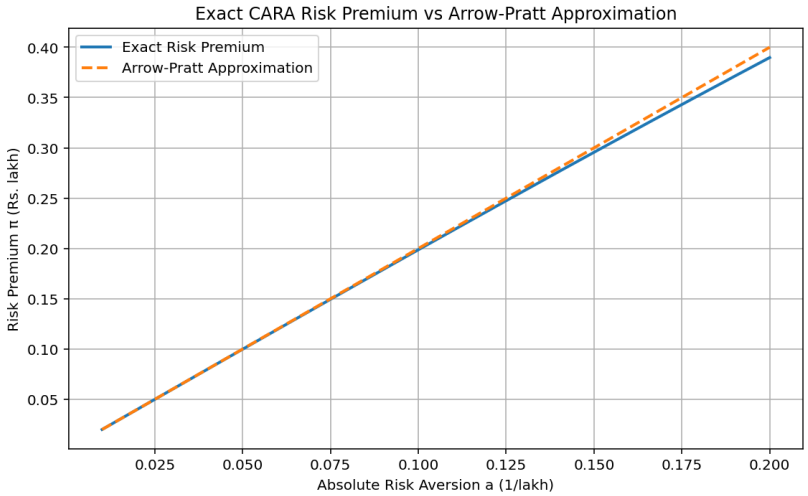


Figure 4: Enter Caption